

MEETS AND JOINS OF GROTHENDIECK TOPOLOGIES IN THE SAMENESS LATTICE

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ABSTRACT. For a small category S , every Grothendieck topology J determines the class W_J of maps of presheaves inverted by sheafification, an element of the complete lattice $\text{Same}(\text{Presh}(S))$ of all two-out-of-three classes. We study how the lattice operations of $\text{Top}(S)$ and of $\text{Same}(\text{Presh}(S))$ interact under $J \mapsto W_J$. Meets are always preserved: $W_{\bigwedge J_i} = \bigcap W_{J_i}$, because the sieves that test local injectivity and surjectivity of a map do not depend on the topology. Joins are preserved up to saturation, $W_{J \vee J'} = \text{Sat}(W_J \vee W_{J'})$, and we prove that they are preserved on the nose for every finite category whose two topologies have an idempotent intersection of ideals — in particular for all finite acyclic categories (finite posets, parallel-arrow categories) and for all but four of the 4623 ordered pairs of idempotent ideals of the monoids of order at most five. The proof for finite monoids is a manipulation of tensor products of bisets: the tower of alternating plus constructions is representable by the tensor products $D_k \otimes \cdots \otimes D_1$ of the ideals, and two factorisation lemmas for idempotent ideals of finite monoids show that these tensor products collapse to $I \otimes I$, the representing object of sheafification for the join, after four plus steps. Joins are *not* preserved in general: on the poset $\mathbb{N}$ the topologies of the even and the odd points have the degenerate topology as join, whereas the join of their samenesses is inverted by the limit of the tower, so it misses the map from the presheaf of final segments to the terminal presheaf. Thus the sameness lattice retains the pro-limit invariant that the subtopos lattice forgets. In every ∞ -topos, finally, the join of a lex modality with the n -truncation is the n -truncated modality, and in simplicial spaces the join of the realization equivalences with the $(n+1)$ -coskeletal equivalences has as its saturation exactly the maps whose realization is an n -equivalence: the identification of Postnikov truncations as joins of sheaf and homotopy equivalences persists at the ∞ -level up to saturation, because a constant simplicial space is $(n+1)$ -coskeletal iff its value is n -truncated. The finite classifications (monoids of order at most five, their idempotent ideals, the tensor products and the factorisation lemmas) are verified exhaustively by computation from the definitions, and the categorical formulas are cross-checked on extensive finite test sets.

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1. INTRODUCTION

A *sameness* on a category C is a class W of morphisms containing the identities, closed under composition and satisfying two-out-of-three; the samenesses form a complete lattice $\text{Same}(C)$ under inclusion [1, 2]. Two classical sources of samenesses are homotopy theory (weak equivalences) and sheaf theory: for a Grothendieck topology J on a small category S the class

$$W_J = \{f \in \text{Presh}(S) : a_J f \text{ is an isomorphism}\}$$

of J -local isomorphisms is a saturated element of $\text{Same}(\text{Presh}(S))$, and $J \mapsto W_J$ is an order-embedding $\text{Top}(S) \hookrightarrow \text{Same}(\text{Presh}(S))$ [2, Thm. 3.1]. This paper asks to what extent this embedding is a lattice homomorphism.

Results.

- (A) *Meets are preserved* (Theorem 3.1): for every family of topologies, $W_{\bigwedge_i J_i} = \bigcap_i W_{J_i}$. The reason is elementary: a map is a J -local isomorphism iff certain sieves attached to the map are J -covering, and these sieves do not depend on J .
- (B) *Joins are preserved up to saturation* (Proposition 4.1): $W_{J \vee J'} = \text{Sat}(W_J \vee W_{J'})$, and $W_J \vee W_{J'} = W_{J \vee J'}$ iff the alternating sheafification $X \rightarrow a_J X \rightarrow a_{J'} a_J X \rightarrow \cdots$ reaches a $(J \vee J')$ -sheaf in finitely many steps for every X (Lemma 4.2).
- (C) *Joins are preserved on finite categories with idempotent intersection* (Theorems 5.1 and 6.6). For a finite category, topologies are idempotent two-sided ideals of morphisms [2, Thm. 5.1], and the join of two topologies is the largest idempotent ideal inside the intersection of their ideals. If that intersection is itself idempotent — always the case for acyclic finite categories, and for 4619 of the 4623 ordered pairs of idempotent ideals of monoids of order ≤ 5 — then $W_J \vee W_{J'} = W_{J \vee J'}$. For acyclic categories the proof is a one-object-at-a-time replacement; for monoids it is a computation with tensor products of bisets: the tower of alternating plus constructions is represented by $D_k \otimes_M \cdots \otimes_M D_1$, and two factorisation lemmas for idempotent ideals of finite monoids (every element factors through an idempotent of the ideal, with the idempotent absorbed on either side) show that these tensor products collapse to $I \otimes_M I$ after four plus steps.
- (D) *Joins are not preserved in general* (Theorem 7.1). On the poset $\mathbb{N}$, the rigid topologies of the even and of the odd points have the degenerate topology as their join, so $W_{J \vee J'}$ is everything; but the limit of the tower inverts both samenesses, hence their join, and does not invert $X \rightarrow 1$ for the presheaf of final segments. The sameness lattice remembers a pro-limit invariant that the subtopos lattice forgets.

- (E) *The ∞ -categorical manifestation* (Theorems 8.1 and 8.3): in every ∞ -topos the join of the sameness of a lex modality L with that of the n -truncation is the sameness of $\tau_{\leq n}L$ — an instance of the join mechanism, since lex localizations commute with truncation; and in simplicial spaces the saturation of the join of the realization equivalences with the $(n+1)$ -coskeletal equivalences is the class of maps whose realization is an n -equivalence, because a constant simplicial space is $(n+1)$ -coskeletal iff its value is n -truncated. Whether that join is already saturated, as it is for simplicial sets [2], is left open.

The finite computations are described in §9; the residual case of (C) for monoids, where the intersection of the two ideals is not idempotent, is isolated as a single combinatorial statement (Conjecture 6.10), verified on all triples of idempotent ideals of monoids of order ≤ 5 ; its profunctor form would settle all finite categories.

Related work. Sheafification, local isomorphisms and the lattice of subtoposes are classical [3, 4]; the correspondence between topologies on a finite category and idempotent two-sided ideals is the finite-sieve form of the Kelly–Lawvere description of essential localizations [5], and the rigidity of topologies on Cauchy-complete categories with finite slices, in particular on finite Cauchy-complete categories, is in [4, 6]. Tensor products of acts over semigroups and the “firm” acts of semigroup Morita theory [7] are the natural home of our bimodule computations; the factorisation lemmas of §6 are of the same flavour as the firmification of acts over factorisable semigroups. Anel, Biedermann, Finster and Joyal [8] describe left-exact localizations of ∞ -topoi by the classes of maps they invert, the *congruences* (strongly saturated classes with finite-limit and base-change stability), and study the lattice of these; there joins are taken at the saturated level. The question here is complementary: whether the join taken *before* saturation, in the lattice of bare two-out-of-three classes, already agrees with the saturated one — Proposition 4.1 is exactly the comparison of the two joins, and Theorems 5.1, 6.6, 6.8 and 7.1 decide it in the cases considered.

2. PRELIMINARIES

2.1. Samenesses and saturation. For a category C let $\text{Same}(C)$ be the complete lattice of samenesses. For $W \in \text{Same}(C)$ an object Z is *W -local* if $\text{Hom}(f, Z)$ is bijective for all $f \in W$; $\text{Loc}(W)$ is the class of W -local objects and $\text{Sat}(W) = \{f : \text{Hom}(f, Z) \text{ bijective for all } Z \in \text{Loc}(W)\}$ the saturation. If L is a reflector onto a full replete subcategory $\mathcal{E}$, then $W_L = L^*(\text{Iso})$ is saturated with $\text{Loc}(W_L) = \mathcal{E}$ [2, Lemma 2.3]. We use throughout:

Lemma 2.1. $\text{Loc}(U \vee V) = \text{Loc}(U) \cap \text{Loc}(V)$ for $U, V \in \text{Same}(C)$.

Proof. The class $\{f : \text{Hom}(f, Z) \text{ bijective}\}$ is a sameness; if it contains U and V it contains their join. $\square$

2.2. Topologies, sheafification, the finite dictionary. For a small category S , $\text{Top}(S)$ denotes the complete lattice of Grothendieck topologies; meets are intersections. a_J is sheafification, the left-exact reflector onto $\text{Sh}_J(S)$, and $W_J := a_J^*(\text{Iso})$ is the class of J -local isomorphisms: $f: X \rightarrow Y$ is J -locally surjective if for every c and $y \in Y(c)$ the sieve $\{h: d \rightarrow c : y \cdot h \in \text{im } f_d\}$ is J -covering, and J -locally injective if for all $x, x' \in X(c)$ with $f(x) = f(x')$ the sieve $\{h: x \cdot h = x' \cdot h\}$ is J -covering [3, III.7].

If every object of S has finitely many sieves (e.g. S finite), topologies correspond, by reverse inclusion, to the *idempotent two-sided ideals* D of S (classes of morphisms closed under pre- and post-composition with $D \circ D = D$): $J_D(c) = \{R \supseteq D(c)\}$ and $D_J(c) = \bigcap J(c)$ [2, Thm. 5.1]. Under this dictionary $J_D \leq J_E$ iff $D \supseteq E$, $J_D \wedge J_E = J_{D \cup E}$, and $J_D \vee J_E = J_I$ where I is the largest idempotent ideal contained in $D \cap E$ (a union of idempotent ideals is idempotent, so this largest ideal exists). We write W_D for W_{J_D} .

2.3. The plus construction and local isomorphisms. For a covering sieve R on c and a presheaf X , $X^{+R}(c) = \text{Hom}(R, X)$; when $J = J_D$ has least covering sieves the plus construction is $X^+ = \text{Hom}(D(-), X)$ and $a_J X = X^{++}$. The unit $X \rightarrow X^+$ is a J -local isomorphism [3, III.5], hence lies in W_J .

3. MEETS ARE PRESERVED

Theorem 3.1. *For any small category S and any family $(J_i)_{i \in I}$ of topologies, $W_{\bigwedge_i J_i} = \bigcap_i W_{J_i}$. Hence $J \mapsto W_J$ preserves arbitrary meets, and $\text{Top}(S)$ is a meet-subsemilattice of $\text{Same}(\text{Presh}(S))$.*

Proof. The meet of topologies is their intersection: a sieve is $(\bigwedge J_i)$ -covering iff it is J_i -covering for every i . A map f is a J -local isomorphism iff the sieves $\{h: y \cdot h \in \text{im } f\}$ and $\{h: x \cdot h = x' \cdot h\}$ attached to f are J -covering; these sieves depend on f only. So f is a $(\bigwedge J_i)$ -local isomorphism iff it is a J_i -local isomorphism for all i . $\square$

4. JOINS: SATURATION AND THE REDUCTION LEMMA

Proposition 4.1. *For topologies J, J' : $W_{J \vee J'} = \text{Sat}(W_J \vee W_{J'})$. In particular $W_J \vee W_{J'} \subseteq W_{J \vee J'}$, with equality iff $W_J \vee W_{J'}$ is saturated.*

Proof. A presheaf is a $(J \vee J')$ -sheaf iff it is a J -sheaf and a J' -sheaf, because the sieves for which a given presheaf and all its pullbacks satisfy the sheaf condition form a topology [4, C2.1]. By Lemma 2.1, $\text{Loc}(W_J \vee W_{J'}) = \text{Sh}_J \cap \text{Sh}_{J'} = \text{Sh}_{J \vee J'} = \text{Loc}(W_{J \vee J'})$, and $W_{J \vee J'}$ is saturated. $\square$

Write $V := W_J \vee W_{J'}$. Two towers will be used: the *alternating sheafification tower* $X \rightarrow a_J X \rightarrow a_{J'} a_J X \rightarrow \dots$ and, when J, J' have least covering sieves, the *alternating plus tower* $X \rightarrow X^{+J} \rightarrow (X^{+J})^{+J'} \rightarrow \dots$; in either case write $\Phi_k X$ for the k -th stage. Every map in either tower is a unit of a_J

or of $a_{J'}$, or a plus-construction map $Y \rightarrow Y^+$, hence lies in W_J or in $W_{J'}$ (§2). The lemma below applies verbatim to both towers.

Lemma 4.2 (reduction). (1) For every map $f: f \in V \iff a_J f \in V \iff \Phi_k f \in V$.
 (2) If $a_{J'} a_J f$ is an isomorphism then $f = g \circ h$ with $h \in W_J$ and $g \in W_{J'}$; in particular $f \in V$.
 (3) If, for one of the two towers, $\Phi_k X$ is a $(J \vee J')$ -sheaf for every X and some $k = k(X)$, then $W_J \vee W_{J'} = W_{J \vee J'}$.

Proof. (1) Naturality $\eta_{X'} f = a_J(f) \eta_X$ with $\eta \in W_J \subseteq V$ and two-out-of-three, twice; the same for each stage of either tower. (2) Let $Z := X' \times_{a_J X'} a_J X$, with $f = g \circ h$ for the induced $h: X \rightarrow Z$ and the projection $g: Z \rightarrow X'$. Then g is a pullback of $a_J f$, which is a J' -local isomorphism when $a_{J'} a_J f$ is iso, and local isomorphisms are pullback-stable; and $Z \rightarrow a_J X$ is a pullback of $\eta_{X'} \in W_J$, so $(Z \rightarrow a_J X) \circ h = \eta_X \in W_J$ gives $h \in W_J$. (3) For $f: X \rightarrow X'$ in $W_{J \vee J'}$ take k working for X and X' ; $\Phi_k f$ is a map between $(J \vee J')$ -sheaves inverted by $a_{J \vee J'}$, hence an isomorphism, so $f \in V$ by (1). See Figure 1. $\square$

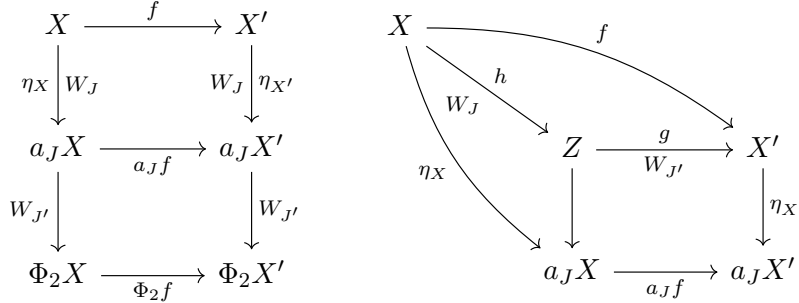


FIGURE 1. Lemma 4.2: the alternating tower (left) moves the question from f to $\Phi_k f$; the pullback trick (right) factors f through a W_J -map and a $W_{J'}$ -map when $a_{J'} a_J f$ is invertible.

Alt text: Left, a ladder of two commutative squares from f down to its double sheafification, with vertical units labelled by the two samenesses. Right, a pullback square of the sheafified map along the unit of the target, with f factored through the pullback object.

5. FINITE ACYCLIC CATEGORIES

Call a finite category *acyclic* if $\text{Hom}(a, b) \neq \emptyset \neq \text{Hom}(b, a)$ implies $a = b$ and the only endomorphisms are identities. Such a category is Cauchy complete (its only idempotents are identities), every topology on it is rigid [4, C2.2.21], [6], and by the dictionary the topologies are the subsets $Y \subseteq$

$\text{Ob } S$: J_Y has ideal $D_Y = \{u: a \rightarrow b : u \text{ factors through some } y \in Y\}$, $\text{Sh}_{J_Y} \simeq \text{Presh}(Y)$ by restriction, $a_{J_Y} = i_* i^*$ for $i: Y \hookrightarrow S$, and $W_Y = \{f : f_y \text{ bijective for all } y \in Y\}$. The join of J_Y and $J_{Y'}$ is $J_{Y \cap Y'}$, and $D_Y \cap D_{Y'} = D_{Y \cap Y'}$ is idempotent.

Theorem 5.1. *For a finite acyclic category S and all $Y, Y' \subseteq \text{Ob } S$, $W_Y \vee W_{Y'} = W_{Y \cap Y'}$. Hence $J \mapsto W_J$ is a lattice embedding $\text{Top}(S) \cong (2^{\text{Ob } S})^{\text{op}} \hookrightarrow \text{Same}(\text{Presh}(S))$.*

Proof. Let $f: X \rightarrow X'$ be bijective on $Y \cap Y'$. Choose a linear extension $p_1, \dots, p_m$ of the preorder $a \preceq b \iff \text{Hom}(a, b) \neq \emptyset$ (predecessors first) and put $S_i = \{p_1, \dots, p_i\}$, a set closed under predecessors. Define $X_i(q) = X'(q)$ for $q \in S_i$ and $X(q)$ otherwise, with restriction along $u: q \rightarrow q'$ given by $X'(u)$ if both objects are in S_i , by $X(u)$ if both are outside, and by $f_q \circ X(u)$ if $q \in S_i, q' \notin S_i$ (the fourth case cannot occur). Functoriality is naturality of f . The maps $\varphi_i: X_{i-1} \rightarrow X_i$ (identity away from p_i , f_{p_i} at p_i) are natural, $f = \varphi_m \circ \dots \circ \varphi_1$, and φ_i is bijective away from p_i . If $p_i \notin Y$ then $\varphi_i \in W_Y$; if $p_i \notin Y'$ then $\varphi_i \in W_{Y'}$; and if $p_i \in Y \cap Y'$ then φ_i is an isomorphism. Hence $f \in W_Y \vee W_{Y'}$. The reverse inclusion is Proposition 4.1, and meets are Theorem 3.1 (with $J_Y \wedge J_{Y'} = J_{Y \cup Y'}$). $\square$

$$\begin{array}{ccc}
 X(q') & \xrightarrow{X(u)} & X(q) \\
 \parallel & & \downarrow f_q \\
 X_i(q') & \xrightarrow{f_q \circ X(u)} & X'(q) = X_i(q)
 \end{array}
 \qquad
 \begin{array}{ccc}
 X(p_i) & \xrightarrow{f_{p_i}} & X'(p_i) \\
 \downarrow X(u) & & \downarrow X'(u) \\
 X(q) & \xrightarrow{f_q} & X'(q)
 \end{array}$$

FIGURE 2. Theorem 5.1: the mixed presheaf X_i (left: the restriction from an unprocessed object q' to a processed object q is $f_q \circ X(u)$) and the naturality square making φ_i a map of presheaves at the newly processed object p_i (right).

Alt text: Two commutative squares. Left: the value of X at an unprocessed object restricts to the value of X at a processed object and then maps by f to the value of X' there. Right: the component of f at the newly processed object commutes with the restriction maps of X and X' .

The acyclicity is what allows the value at a single object to be replaced; with cycles, an object in $D \setminus D'$ and one in $D' \setminus D$ may be mutually reachable, and the replacement must be organised differently (§6).

6. FINITE MONOIDS: THE TENSOR TOWER

Let M be a finite monoid, $\text{Presh}(M)$ the category of right M -sets, and D, E idempotent two-sided ideals, with J_D, J_E the corresponding topologies. Right ideals are the sieves; D and E are the least covering sieves. For a right

M -set Z and an M - M -biset P we have the tensor $Z \otimes_M P$ (quotient of $Z \times P$ by $(zm, p) \sim (z, mp)$, a right M -set) and the adjunction

$$(1) \quad \text{Hom}_M(Z \otimes_M P, X) \cong \text{Hom}_M(Z, \text{Hom}_M(P, X)),$$

where $\text{Hom}_M(P, X)$ carries the right action $(\varphi \cdot m)(p) = \varphi(mp)$. Consequently the alternating plus tower (one plus construction per factor; sheafification is two consecutive plus constructions) is representable:

$$\Phi_k X := ((X^{+D})^{+E})^{+D \cdots} = \text{Hom}_M(T_k, X),$$

$$T_k := D_k \otimes_M \cdots \otimes_M D_1, \quad D_1 = D, \ D_2 = E, \ D_3 = D, \ \dots,$$

with the unit $\Phi_{k-1} X \rightarrow \Phi_k X$ induced by the multiplication $\mu_k: T_k = D_k \otimes T_{k-1} \rightarrow T_{k-1}$. Each unit is in W_D or W_E . For an idempotent ideal I , $a_I X = \text{Hom}_M(I \otimes_M I, X)$.

We write $[x_k, \dots, x_1]$ for the class of a tuple in T_k ; the relations are $[\dots, am, b, \dots] = [\dots, a, mb, \dots]$ across each tensor sign, subject to every entry staying in its slot ideal.

6.1. Two factorisation facts.

Lemma 6.1. *Let Y be an idempotent two-sided ideal of a finite monoid. Every $y \in Y$ admits factorisations*

$$(F1) \quad y = y'gy'' \quad \text{with } g = g^2 \in Y, \ y'g = y', \ y', y'' \in Y;$$

$$(F2) \quad y = y'gy'' \quad \text{with } g = g^2 \in Y, \ gy'' = y'', \ y', y'' \in Y.$$

Proof. Write $y = y_1 \cdots y_N$ with $y_i \in Y$ and $N > |M| + 2$. Among the prefixes $p_j = y_1 \cdots y_j$ ($1 \leq j \leq |M| + 1$) two coincide, $p_i = p_j$ with $i < j$, so $p_i = p_i m$ for $m = y_{i+1} \cdots y_j \in Y$, hence $p_i = p_i m^r$ for every $r \geq 1$; since M is finite some power $g = m^r$ is idempotent, $g \in Y$ because Y is closed under products, and $p_i = p_i g$. Then $y = p_i g (y_{j+1} \cdots y_N)$ is (F1). (F2) is the same argument on suffixes. $\square$

6.2. Merging lemmas.

Lemma 6.2 (P). *Let X, Y, Z be two-sided ideals with Y idempotent. In $X \otimes Y \otimes Z$, for $c \in Y \cdot Z$ the class $[x, y, c]$ depends only on (xy, c) .*

Proof. Fix a rule writing $c = y_1 z_1$ ($y_1 \in Y, z_1 \in Z$) and $y_1 = y'gy''$ by (F1). Moving $y'g$ to the left (the remaining $y''z_1$ lies in Z) and then yy' to the left (the remaining g lies in Y):

$$[x, y, c] = [x, y y'g, y''z_1] = [xyy', g, y''z_1],$$

which depends on xy and on the data (y', g, y'', z_1) determined by c . $\square$

Lemma 6.3 (P'). *With Y idempotent, for $c \in X \cdot Y$ the class $[c, y, z] \in X \otimes Y \otimes Z$ depends only on (c, yz) . (Mirror image, using (F2).)*

Lemma 6.4 (absorption). *If D_j is idempotent, $[\dots, x_j, x_{j-1}, \dots] = [\dots, x'_j, x''_j x_{j-1}, \dots]$ for any factorisation $x_j = x'_j x''_j$ in D_j ; so an interior slot may be assumed to lie in the product of its ideal with a neighbouring ideal.*

Lemma 6.5 (merging). *Let $X \otimes Y \otimes Z \otimes \dots$ be a tensor of ideals with Y idempotent. If $[x, y, c, \dots]$ depends only on $(xy, c, \dots)$ for all tuples, then the multiplication $X \otimes Y \otimes Z \otimes \dots \rightarrow XY \otimes Z \otimes \dots$ is a bijection (and likewise for merging a later pair under the hypothesis of Lemma P').*

Proof. The map is surjective. Define a section on tuples of the target by $L[w, c, \dots] := [x, y, c, \dots]$ for any factorisation $w = xy$; it is well defined by hypothesis, and it respects the relations of the target: for $[wm, c] \sim [w, mc]$ use the factorisation $wm = x(ym)$; for $[w, cm] \sim [w, c, m]$ nothing changes in the first two slots. So L descends to the target, and $L \circ \mu = \text{id}$. $\square$

6.3. The theorem.

Theorem 6.6. *Let M be a finite monoid and D, E idempotent two-sided ideals such that $I := D \cap E$ is idempotent (so $J_D \vee J_E = J_I$). Then the canonical multiplication maps give isomorphisms of bisets $T_k \cong I \otimes_M I$ for all $k \geq 4$. Consequently $\Phi_4 X \cong a_I X$ for every M -set X : the alternating plus tower reaches the J_I -sheafification after four plus steps, and*

$$W_D \vee W_E = W_{D \cap E} \quad \text{in } \text{Same}(\text{Presh}(M)).$$

Proof. $DE \subseteq I$ and $ED \subseteq I$, $I \subseteq D$, $I \subseteq E$, and every product of two or more alternating factors D, E lies between $I = I^2$ and $D \cap E = I$, hence equals I . Write $T_k = [x_1, \dots, x_k]$ with slot ideals $D_k, D_{k-1}, \dots, D_1$ from left to right.

Step 1: merge slots 1, 2. Let t, t' be tuples differing only in slots 1, 2, with equal products $x_1 x_2 = x'_1 x'_2$. Absorb slot 4 into slot 3 in both, with the same factorisation $x_4 = pq$ (Lemma 6.4; here $k \geq 4$ is used): slot 3 becomes $x_3 p \in D_{k-2} D_{k-3} = I \subseteq D_{k-1} D_{k-2}$. Lemma P for the triple of slots 1, 2, 3 shows that the two classes agree. By Lemma 6.5 the multiplication $T_k \rightarrow (D_k D_{k-1}) \otimes D_{k-2} \otimes \dots \otimes D_1 = I \otimes D_{k-2} \otimes \dots \otimes D_1$ is a bijection.

Step 2: merge slots 2, 3 repeatedly. In $I \otimes Y \otimes Z \otimes \dots$ with Y idempotent, every element c of the first slot lies in $I = I \cdot I \subseteq I \cdot Y$, so Lemma P' applies to every tuple and Lemma 6.5 gives a bijection onto $I \otimes YZ \otimes \dots$. Starting from $I \otimes D_{k-2} \otimes D_{k-3} \otimes \dots \otimes D_1$, the first such merge produces $I \otimes I \otimes D_{k-4} \otimes \dots$ ($D_{k-2} D_{k-3} = I$), and since I is idempotent the merge can be repeated ($I \cdot D_j = I$) until two slots remain: $T_k \cong I \otimes I$. All maps are multiplications, so the composite $T_k \rightarrow I \otimes I$ is the canonical one and is an isomorphism of M - M -bisets (Figure 3).

Therefore $\Phi_k X = \text{Hom}_M(T_k, X) \cong \text{Hom}_M(I \otimes I, X) = a_I X$ as M -sets, so $\Phi_k X$ is an I -sheaf; and $X \rightarrow \Phi_k X$ is a composite of units in $W_D \cup W_E \subseteq W_I$, so $\Phi_k X \cong a_I X$ under X and the unit $\eta_X : X \rightarrow a_I X$ lies in $W_D \vee W_E$. For $f \in W_I$ the naturality square $\eta_{X'} f = a_I(f) \eta_X$ with $a_I f$ invertible gives $f \in W_D \vee W_E$. The reverse inclusion is Proposition 4.1. $\square$

$$E \otimes D \otimes E \otimes D \xrightarrow{P, (1,2)} I \otimes E \otimes D \xrightarrow{P', (2,3)} I \otimes I$$

FIGURE 3. The collapse of T_4 in Theorem 6.6: the first merge uses Lemma P after absorbing the last slot into the third, the second uses Lemma P' with the first slot already in I . Both arrows are multiplication maps and both are bijections.

Alt text: A chain of two multiplication maps: the four-fold tensor of E, D, E, D merges its first pair into I , then the three-fold tensor merges its last pair, ending at I tensor I .

6.4. Finite categories. Let S be a finite category. A *two-sided ideal* D of S is a class of morphisms closed under pre- and post-composition, i.e. a sub-profunctor $D \subseteq \text{Hom}_S$; it is *idempotent* if $D \circ D = D$ (every member factors as a composite of two members). For ideals (or, more generally, sub-profunctors of Hom_S) P, Q the composite profunctor is

$$(Q \otimes_S P)(a, c) = \left(\coprod_b Q(b, c) \times P(a, b) \right) / ((q \circ m, p) \sim (q, m \circ p)),$$

an S - S -profunctor, and for a presheaf X and a profunctor P the presheaf $\text{Hom}(P, X)(c) := \text{Hom}_{\text{Presh}(S)}(P(-, c), X)$ satisfies $\text{Hom}(Q \otimes_S P, X) \cong \text{Hom}(Q, \text{Hom}(P, X))$, the profunctor form of (1). If $J = J_D$ then $X^+ = \text{Hom}(D, X)$, $a_J X = X^{++}$, and the alternating plus tower is $\Phi_k X = \text{Hom}(T_k, X)$ with $T_k = D_k \otimes_S \cdots \otimes_S D_1$. Elements of $T_k(a, c)$ are classes $[x_k, \dots, x_1]$ of composable chains $a \xrightarrow{x_1} \cdots \xrightarrow{x_k} c$ with $x_i \in D_i$, modulo the relations $[\dots, x_{i+1} \circ m, x_i, \dots] = [\dots, x_{i+1}, m \circ x_i, \dots]$ with all entries in their slot ideals.

Lemma 6.7. *Let Y be an idempotent ideal of the finite category S . Every $y \in Y(a, c)$ admits factorisations of both types: $y = y' \circ g \circ y''$ with $g \in Y$ an idempotent endomorphism, $y', y'' \in Y$, and $y' \circ g = y'$ (F1); and likewise one with $g \circ y'' = y''$ (F2).*

Proof. Write $y = y_N \circ \cdots \circ y_1$ with $y_i \in Y$ and N larger than the number of morphisms of S plus two. The partial composites $p_j = y_j \circ \cdots \circ y_1$ all have domain a , so two coincide, $p_i = p_j$ with $i < j$; then $p_i = m \circ p_i$ for the endomorphism $m = y_j \circ \cdots \circ y_{i+1} \in Y$ of $\text{cod}(p_i)$, some power $g = m^r$ is idempotent, $g \in Y$, and $g \circ p_i = p_i$. Then $y = (y_N \circ \cdots \circ y_{j+1}) \circ g \circ p_i$ is (F2); (F1) is the same argument with the partial composites $y_N \circ \cdots \circ y_j$ of common codomain c . $\square$

Theorem 6.8. *Let S be a finite category and J_D, J_E topologies whose ideals have an idempotent intersection $I = D \cap E$ (so $J_D \vee J_E = J_I$). Then $T_k \cong I \otimes_S I$ canonically for $k \geq 4$, and $W_D \vee W_E = W_{D \cap E}$ in $\text{Same}(\text{Presh}(S))$.*

Proof. Lemmas P, P', 6.4 and 6.5 are statements about classes of composable chains and the relations across tensor signs; their proofs use only the relations, the closure of ideals under composition on both sides, and Lemma 6.1,

which Lemma 6.7 replaces. (In Lemma P, for $c \in (Y \circ Z)(a, b)$ write $c = y_1 \circ z_1$ and $y_1 = y' \circ g \circ y''$ by (F1); then $[x, y, c] = [x, y \circ y' \circ g, y'' \circ z_1] = [x \circ y \circ y', g, y'' \circ z_1]$, each step legitimate because the remaining entry lies in its slot ideal.) The proof of Theorem 6.6 is then repeated word for word: after absorbing slot 4 into slot 3, Lemma P merges slots 1, 2 into $D_k D_{k-1} = I$; Lemma P' then merges slots 2, 3 repeatedly, since the first slot lies in $I = I \circ I \subseteq I \circ Y$; every alternating product of ≥ 2 factors equals I . Hence $\Phi_k X = \text{Hom}(T_k, X) \cong \text{Hom}(I \otimes I, X) = a_I X$ for $k \geq 4$, and the last paragraph of the proof of Theorem 6.6 gives $W_D \vee W_E = W_I$. $\square$

Theorem 5.1 is the special case of acyclic S , where every intersection of ideals is idempotent (the ideals are generated by the identities of subsets of objects), with a more elementary proof.

Example 6.9 (semilattice monoids). For a finite meet-semilattice L as a monoid, ideals are down-sets and all are idempotent, so Theorem 6.6 applies to every pair; for principal down-sets $D = \downarrow e$, $E = \downarrow e'$ the unit chain is explicit, $X \rightarrow X \cdot e \rightarrow X \cdot e \cdot e' = X \cdot (e \wedge e')$, two steps.

6.5. The residual case. The hypothesis of Theorem 6.6 can fail: among the 4623 ordered pairs of idempotent ideals of the 273 monoids of order ≤ 5 there are four pairs (in two monoids of order 5) with $D \cap E$ not idempotent; there $DE \neq ED$ and the join corresponds to the smaller ideal $I' = (DE)^m$. In these cases the alternating tower still stabilises, to $I' \otimes I'$, in at most four steps, and the collapse is explicit (an element $2 \in D \cap E$ with $2 = 3 \cdot 2 = 2 \cdot 4$, $3 \in D$ and $4 \in E$ idempotent, $2 \cdot 3 = 0$). The missing ingredient is the following statement. The unrestricted version (allowing $Z = M$) fails in exactly six of the 10 235 triples of idempotent ideals of monoids of order ≤ 5 , all with $Z = M$, where it reduces to the false two-fold statement $X \otimes Y \rightarrow XY$; under the hypothesis $Z \neq M$ no failure occurs.

Conjecture 6.10 (three-fold multiplicativity). *For idempotent two-sided ideals $X, Y, Z \neq M$ of a finite monoid, the class $[x, y, c] \in X \otimes Y \otimes Z$ depends only on (xy, c) for all $c \in Z$; equivalently $X \otimes Y \otimes Z \rightarrow XY \otimes Z$ is a bijection.*

By Lemma 6.5 the conjecture implies $T_k \cong I' \otimes I'$ for k large and hence $W_D \vee W_E = W_{J_D \vee J_E}$ for all finite monoids. For a finite category the corresponding statement is the profunctor form of the conjecture — for idempotent ideals X, Y, Z of S with Z not containing any identity, the class $[x, y, c] \in X \otimes_S Y \otimes_S Z$ depends only on $(x \circ y, c)$ — which does not follow from the monoid case and is left as a separate question; with it, the argument of §6.4 would give join preservation for all finite categories.

7. JOINS ARE NOT PRESERVED IN GENERAL

Theorem 7.1. *Let $S = (\mathbb{N}, \leq)$ and let $J_{\text{ev}}, J_{\text{od}}$ be the topologies of the ideals $D_Y = \{n \rightarrow m : \exists y \in Y, n \leq y \leq m\}$ for Y the even and the odd*

numbers. Then $J_{\text{ev}} \vee J_{\text{od}}$ is the degenerate topology, so $W_{J_{\text{ev}} \vee J_{\text{od}}} = \text{Mor}$, but $W_{J_{\text{ev}}} \vee W_{J_{\text{od}}} \subsetneq \text{Mor}$: it is contained in the class of maps inverted by $\lim_{\mathbb{N}^{\text{op}}} : \text{Presh}(\mathbb{N}) \rightarrow \mathbf{Set}$.

Proof. D_Y is an idempotent two-sided ideal, so J_Y is a topology [2, Thm. 5.1] (that direction needs no finiteness); it is rigid with irreducible objects Y , $\text{Sh}_{J_Y} \simeq \text{Presh}(Y)$ by restriction, $(a_Y X)(m) = X(\max\{y \in Y : y \leq m\})$ when the maximum exists and $(a_Y X)(m) = 1$ otherwise (for Y the odd numbers and $m = 0$ the indexing category is empty), and $W_Y = \{f : f_y \text{ bijective for } y \in Y\}$. The join J is degenerate: we show $\emptyset \in J(m)$ by induction on m . For $m = 0$ the least J_{od} -covering sieve $D_{\text{od}}(0)$ is empty. For $m > 0$ let $R = D_{\text{od}}(m)$ if m is even and $R = D_{\text{ev}}(m)$ if m is odd; R is J -covering, and every arrow $n \rightarrow m$ in R has $n \leq y < m$ for some y of the other parity, so $n < m$ and $\emptyset \in J(n)$ by induction; transitivity gives $\emptyset \in J(m)$. Now $\Lambda := \lim_{\mathbb{N}^{\text{op}}}$ inverts W_{ev} and W_{od} , since evens and odds are cofinal in the tower, hence inverts their join. For $X(n) = \{x \in \mathbb{N} : x \geq n\}$ with inclusions, $\Lambda X = \emptyset$, so $\Lambda(X \rightarrow 1)$ is not bijective and $X \rightarrow 1$ is not in the join. $\square$

The mechanism is the failure of objectwise stabilisation in Lemma 4.2: here $(\Phi_k X)(m) \approx X(m - k)$ shifts by one at each step and never reaches the terminal sheaf. $\text{Same}(\text{Presh}(\mathbb{N}))$ retains the pro-limit invariant Λ that the lattice of subtoposes forgets; the saturation of the join is everything, the join itself is a proper sameness.

$$\begin{array}{ccccccc} \cdots & \longrightarrow & X(3) & \longrightarrow & X(2) & \longrightarrow & X(1) & \longrightarrow & X(0) \\ & & \parallel & & \downarrow & & \parallel & & \downarrow \\ \cdots & \longrightarrow & X(3) & \longrightarrow & X(2) & \longrightarrow & X(1) & \longrightarrow & * \end{array}$$

FIGURE 4. Theorem 7.1: a map bijective on the odd points (vertical equalities) may change the even values arbitrarily; iterating with the even points shifts the tower by one step at a time, and the limit Λ never changes.

Alt text: Two towers of sets indexed by the natural numbers, connected by vertical maps that are equalities at the odd indices and arbitrary at the even indices.

8. THE ∞ -CATEGORICAL MANIFESTATION

Samenesses of an ∞ -category are classes of morphisms containing the equivalences, closed under composition and two-out-of-three; the join theorem of [2, Thm. 4.1] and its corollary for two reflectors hold verbatim, as does Lemma 2.1. The results of this section are instances of that mechanism in the well-developed theory of localizations of ∞ -topoi [9]; we record them because they say precisely what survives of the sheaf/homotopy comparison at the ∞ -level.

Let $\mathcal{E}$ be an ∞ -topos, L the reflector of a subtopos (an accessible left-exact localization; a lex modality), and $\tau_{\leq n}$ the n -truncation reflector.

Theorem 8.1. *In $\text{Same}(\mathcal{E})$: $W_L \vee W_{\tau_{\leq n}} = W_{\tau_{\leq n}L} = W_{L\tau_{\leq n}}$, the class of maps that become equivalences after passing to the n -truncated L -sheaf; its local objects are the n -truncated L -local objects, and $\tau_{\leq n}L$ is again a reflective subuniverse. If the subtopos of L is hypercomplete, $\bigcap_n (W_L \vee W_{\tau_{\leq n}}) = W_L$.*

Proof. By [2, Cor. 4.2] it suffices that $\tau_{\leq n}L$ inverts $W_{\tau_{\leq n}}$. A left-exact left adjoint between presentable ∞ -categories commutes with truncation, $L\tau_{\leq n} \simeq \tau_{\leq n}L$ [9, 5.5.6.28], so $\tau_{\leq n}Lg \simeq L\tau_{\leq n}g$ is an equivalence whenever $\tau_{\leq n}g$ is. The intersection consists of the maps f with Lf ∞ -connective, which are the L -equivalences when the subtopos is hypercomplete. $\square$

Now let $\mathcal{P} = \text{Fun}(\Delta^{\text{op}}, \mathcal{S})$ be the ∞ -category of simplicial spaces, $|-| : \mathcal{P} \rightarrow \mathcal{S}$ the colimit (geometric realization), left adjoint to the fully faithful constant functor c (fully faithful because Δ^{op} has the initial object $[0]$, so its nerve is contractible and colimits of constant diagrams are trivial), and cosk_{n+1} the $(n+1)$ -coskeleton, the reflector onto the $(n+1)$ -coskeletal objects. Put $W_{|-|} := \{f : |f| \text{ an equivalence}\}$, $W_{\text{cosk}_{n+1}} := \{f : \text{cosk}_{n+1}f \text{ an equivalence}\}$ and $W_{\tau_{\leq n}|-|} := \{f : \tau_{\leq n}|f| \text{ an equivalence}\}$.

Lemma 8.2. *For $K \in \mathcal{S}$ and $n \geq 0$, the constant simplicial space cK is $(n+1)$ -coskeletal iff K is n -truncated.*

Proof. $(\text{cosk}_{n+1}cK)_m = \text{Map}_{\mathcal{P}}(\text{sk}_{n+1}\Delta^m, cK) = \text{Map}_{\mathcal{S}}(|\text{sk}_{n+1}\Delta^m|, K)$, and the unit $cK \rightarrow \text{cosk}_{n+1}cK$ is, at level m , the inclusion of the constant maps. The space $|\text{sk}_{n+1}\Delta^m|$ is contractible for $m \leq n+1$ and n -connected for $m \geq n+2$ (the $(n+1)$ -skeleton of a contractible CW complex), with $|\text{sk}_{n+1}\Delta^{n+2}| = |\partial\Delta^{n+2}| \simeq S^{n+1}$. If K is n -truncated then $\text{Map}(A, K) \simeq \text{Map}(\tau_{\leq n}A, K) \simeq K$ for every n -connected A , so cK is $(n+1)$ -coskeletal. Conversely, at level $n+2$ the unit is $K \rightarrow \text{Map}(S^{n+1}, K)$, whose fibre over the component of x is $\Omega^{n+1}(K, x)$; it is an equivalence iff $\pi_i(K, x) = 0$ for all $i \geq n+1$ and all x , i.e. iff K is n -truncated. $\square$

Theorem 8.3. *In $\text{Same}(\mathcal{P})$, for every $n \geq 0$,*

$$\text{Sat}(W_{|-|} \vee W_{\text{cosk}_{n+1}}) = W_{\tau_{\leq n}|-|},$$

and the local objects of the join are the constant n -truncated simplicial spaces.

Proof. $W_{|-|}$ and $W_{\text{cosk}_{n+1}}$ are the classes inverted by the reflectors $c|-|$ and cosk_{n+1} , so they are saturated with local objects the constant and the $(n+1)$ -coskeletal objects respectively. By Lemma 2.1 and Lemma 8.2 the local objects of the join are the constant n -truncated objects $c(\mathcal{S}_{\leq n})$. This is a reflective subcategory with reflector $c \circ \tau_{\leq n} \circ |-|$ (reflect into constants, then truncate the value), whose inverted class is $W_{\tau_{\leq n}|-|}$; it is saturated with the same local objects, hence equals the saturation of the join. $\square$

Thus the theorem “Postnikov truncation = homotopy $\vee$ coskeleton” of [2] persists at the ∞ -level up to saturation. For simplicial sets the join is already saturated, the proof there using the Kan replacement inside the homotopy sameness; whether $W_{|-|} \vee W_{\text{cosk}_{n+1}}$ is saturated in simplicial spaces we leave open. (For $n = -1$ the same statements hold with “ (-1) -truncated” meaning empty or contractible.)

9. COMPUTATIONAL RECORD

The classifications below (monoids, ideals, tensor products, factorisation lemmas) are exhaustive within the stated ranges; the tests on presheaves and M -sets are cross-checks on finite test sets, not exhaustive (`scripts/comparison/`, Python 3, standard library):

- `topjoin_poset.py`: for all posets on ≤ 4 points and 25 random finite presheaves each, the alternating sheafification reaches a $(Y \cap Y')$ -sheaf in at most $n + 1$ steps ($> 110\,000$ cases).
- `monoids_small.py`: backtracking enumeration of monoids of order ≤ 5 (1, 2, 7, 35, 228, matching the known counts).
- `topjoin_monoid.py`: on the 23 monoids of order ≤ 5 with incomparable idempotent ideals, the alternating chain stabilises in ≤ 3 steps on all cyclic M -sets of size ≤ 4 , and Theorem 3.1 was checked on 12 442 equivariant maps.
- `tensor_tower.py`, `tensor_multi.py`: the towers T_k for all 66 incomparable pairs stabilise by $k = 4$ to the size of $I \otimes I$ (resp. $I' \otimes I'$); $E \otimes D \cong E \cap D$ and $I \otimes I \cong I$ fail in general.
- `lemmaP_check.py`: Lemmas P and P' on all 4245 triples of proper idempotent ideals; `tensor_assoc.py`: the unrestricted form of Conjecture 6.10 on all 10 235 triples — six failures, all with $Z = M$, none under the hypothesis $Z \neq M$.

10. CONCLUSION

Under $J \mapsto W_J$ the lattice of topologies sits inside the sameness lattice as a meet-subsemilattice, and as a sublattice whenever the category is finite and the two ideals intersect idempotently. The counterexample exhibited here is genuinely infinite and is detected by a pro-limit; whether join preservation can fail for a finite category remains open; for finite monoids it is reduced to a single statement about tensor products of idempotent ideals (Conjecture 6.10), and for finite categories to its profunctor form.

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